北京郵電大學

Queen Mary University of London

For examiners' use only

1	
2	
3	
4	
5	
6	
7	
8	
Total	

EBU6501 B

Joint Programme Examinations 2023/24

EBU6501 Middleware

Paper B

Time allowed 2 hours

Answer ALL Questions

Complete the information below about yourself very carefully.

QM student number										
BUPT student number										
Class number										

NOT allowed: Any Electronic devices including electronic calculators and electronic dictionaries.

INSTRUCTIONS

1. **You must NOT take answer books, used or unused, from the examination room.**
2. Write only with a black or blue pen **and in English**.
3. Do all rough work in the answer book – **do not tear out any pages**.
4. If you use Supplementary Answer Books, tie them to the end of this book.
5. Write clearly and legibly.
6. **Read the instructions on the inside cover.**

Examiners

Dr Gokop Goteng, Dr Ethan Lau

Copyright © Beijing University of Posts and Telecommunications & © Queen Mary University of London 2023

Filename: 2324_EBU6501_B No answer book required

Instructions

Before the start of the examination

1) Place your BUPT and QM student cards on the corner of your desk so that your picture is visible.

2) Put all bags, coats and other belongings at the back/front of the room. All small items in your pockets, including wallets, mobile phones and other electronic devices must be **placed in your bag in advance**. **Possession of mobile phones, electronic devices and unauthorised materials is an offence.**

3) Please ensure your mobile phone is switched off and that no alarm will sound during the exam. **A mobile phone causing a disruption is also an assessment offence.**

4) Do not turn over your question paper or begin writing until told to do.

During the examination

1) You must not communicate with or copy from another student.

2) If you require any assistance or wish to leave the examination room for any reason, please raise your hand to attract the attention of the invigilator.

3) If you finish the examination early you may leave, but not in the first 30 minutes or the last 10 minutes.

4) For 2 hour examinations you may **not** leave temporarily.

5) For examinations longer than 2 hours you **may** leave temporarily but not in the first 2 hours or the last 30 minutes.

At the end of the examination

1) You must stop writing immediately – **if you continue writing after being told to stop, that is an assessment offence.**

2) Remain in your seat until you are told you may leave.

Question 1

a) Describe how the objects **`MessageProducer`** and **`MessageConsumer`** communicates using the concept of Message-oriented-middleware (MOM). Draw the MOM architecture diagram to aid the explanation.

(5 marks)

	Do not write in this column
	5 marks

b) Apache ActiveMQ is an example of the Message Broker for the Java Message Service (JMS) application. It is used in publishing messages to a ***Topic*** or ***Queue***.

(15 marks)

i) The Figure 1 (a-b) show the type of the JMS messaging model. Denote the type of the messaging model used in a) and b). Then, explain the difference of the two messaging models.

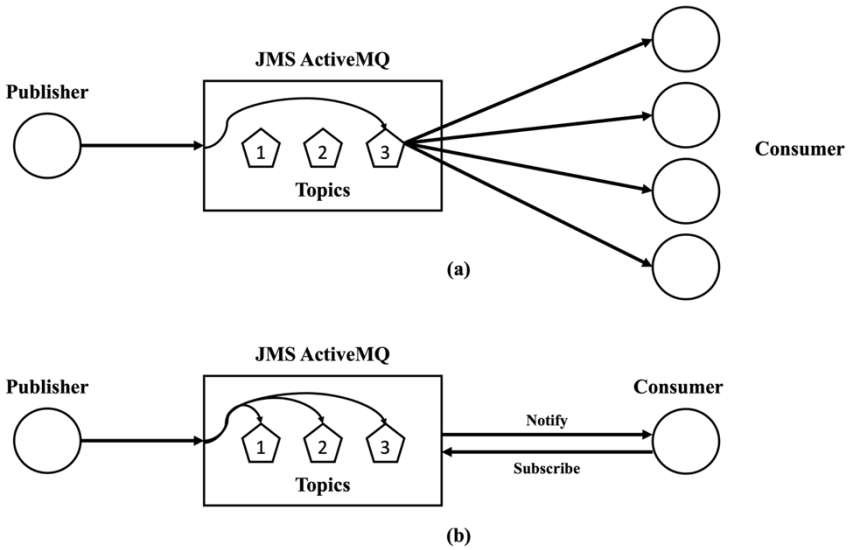


Figure 1 (a-b).

[6 marks]

ii) In your own words, describe the importance of the ***Sessions*** in the context of JMS Application Programming Interface (API) model.

[3 marks]

iii) What are the TWO important objects in JMS that are maintained administratively? Justify your answer accordingly.

[6 marks]

	Do not write in this column
	15 marks

c) The Inversion of Control (IoC) is a principle in the software engineering of the Spring Framework. It is most often used in the context of object-oriented programming such as Java.

(5 marks)

i) List out the TWO examples of the configuration metadata.

[2 marks]

ii) The Application Context is Spring's advanced container. Figure 2 shows a Spring's Application Context by instantiating the **`ApplicationContext`** interface that loads the .xml configurations from a classpath of "**info.xml**". Finally, the bean with the bean ID = **`beanInfo`** is obtained from the **`beanInfo`** class. Complete the missing information [i], [ii] and [iii] correspondingly.

```
public static void main(String[] args) {

    SpringApplication.run(FirstSpringApplication.class, args);
    ClassPathXmlApplicationContext context=new [i]("[ii]");
    beanInfo sb=(beanInfo)context.[iii] ("beanInfo");
    sb.getName();
```

Figure 2.

[3 marks]

	Do not write in this column
	5 marks

Question marking $= \frac{}{5} + \frac{}{15} + \frac{}{5} = \frac{}{25}$

Question 2

a) Examine the Tomcat directory structure in Figure 3 below carefully.

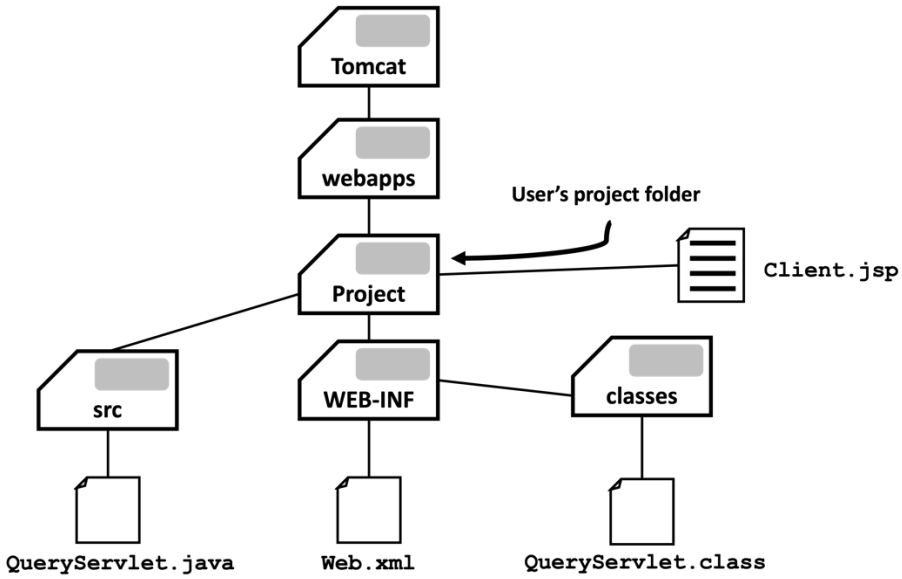


Figure 3.

(5 marks)

i) Define the term given to the file "**web.xml**" and explain why it is needed in the Tomcat Container.
[3 marks]

iii) Provide the correct URL using HTTP protocol and appropriate port number to access the web page "**`client.jsp`**" if you are running the system installed on your local machine.
[2 marks]

	Do not write in this column
	5 marks

b) The servlet class **`QueryServlet`** with servlet name **`Servlet1`** from Figure 4 is implemented in the Deployment Descriptor (DD) and the servlet class is in the directory **`com/query`** that extends **`HttpServlet`** class in Figure 3. The URL pattern is **`/result`**. Write servlet declaration in your DD for this servlet showing the complete package name of the class including the servlet-mapping and the URL in the DD.

```
public class QueryServlet extends HttpServlet {
      public void ModelDatabase() {
      System.out.print("<Query response result>");
      }
}
```

Figure 4.

(7 marks)

	Do not write in this column
	7 marks

c) This question is about Java threads.

(13 marks)

i) Maintaining their order, state and explain the main FIVE life cycle states of a Java thread.

[6 marks]

ii) Figure 5 shows the Java class **`MyAccumulator`** where the method **getNextValue()** is not thread-safe as it is accessible by multiple threads. Improve the code in Figure 5 by acquiring an intrinsic lock to ensure the operation is thread-safe and atomic. Include the **`main()`** method to display the output result.

```
public class MyAccumulator {
  private int value = 50;
  public int getNextValue(){
          return value++;
``
```

Figure 5.

[7 marks]

	Do not write in this column

		13 marks

Question marking $= \frac{}{5} + \frac{}{7} + \frac{}{13} = \frac{}{25}$

Question 3

a) Describe how ***Apache Kafka Streams*** simplifies middleware application development that enables data locality, elasticity, scalability, high performance and fault tolerance system.

[16 marks]

	Do not write in this column
	16 marks

b) Describe ***Injection Flaws*** and ***Cross Site Request Forgery (CSRF)*** security problems and how to prevent each of them.

[7 marks]

	Do not write in this column
	7 marks

c) Describe ONE IoT constrained protocol that is used at the application layer for sensors.

[2 marks]

	Do not write in this column
	2 marks

Question marking: $\frac{}{16} + \frac{}{7} + \frac{}{2} = \frac{}{25}$

Question 4

a) A company intends to use a secure application programming interface (API) registry management service that allows bundles to be remotely managed for developing middleware using Java technology. Describe THREE features in OSGi (Open Service Gateway Initiative) that can meet these requirements.

[6 marks]

	Do not write in this column
	6 marks

b) A company wants to put in place a security system that ensures secure way for users to login, have access to the computer system and use the services. In addition, it wants to ensure that data is not seen by unauthorised users and that data cannot be manipulated while stored on the system or in transit. Discuss FOUR ways that you can help this company to achieve these objectives.

[8 marks]

	Do not write in this column

	8 marks

c) Describe the FOUR steps in using the Uniform Resource Locator (URL) to write and implement a Java networking program.

[11 marks]

	Do not write in this column
	11 marks

Question marking $\frac{}{6} + \frac{}{8} + \frac{}{11} = \frac{}{25}$

Do not write in this column

2023-2024

Rough Working

Page 15 of 17

Do not write in this column

2023-2024

Rough Working

Page 16 of 17

Do not write in this column

2023-2024
Rough Working
Page 17 of 17